

HAT-P-36b

R-1890

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_230601 · created 2026-08-03T12:47:42+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460096.7526 +/- 0.0033 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.153 +/- 0.037
Transit depth (Rp/Rs)^2	2.35 +/- 1.12 [%]
Orbital Inclination (inc)	89.46 +/- 2.13
Airmass coefficient 1 (a1)	1.375 +/- 0.012
Airmass coefficient 2 (a2)	-0.204 +/- 0.024
Scatter in the residuals of the lightcurve fit is	0.87 %
Best Comparison Star	None
Optimal Aperture	3.44
Optimal Annulus	15.23
Transit Duration (day)	0.0673 +/- 0.0037

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.153 $\pm$ 0.037 vs 0.126 (deviation 0.73 σ)
Duration fit vs published	0.0673 d vs 0.0925 d (deviation 6.81 σ)
Mid-time O-C	12.4 min
Depth SNR	4.1
Residual scatter	0.87 %

Light curve

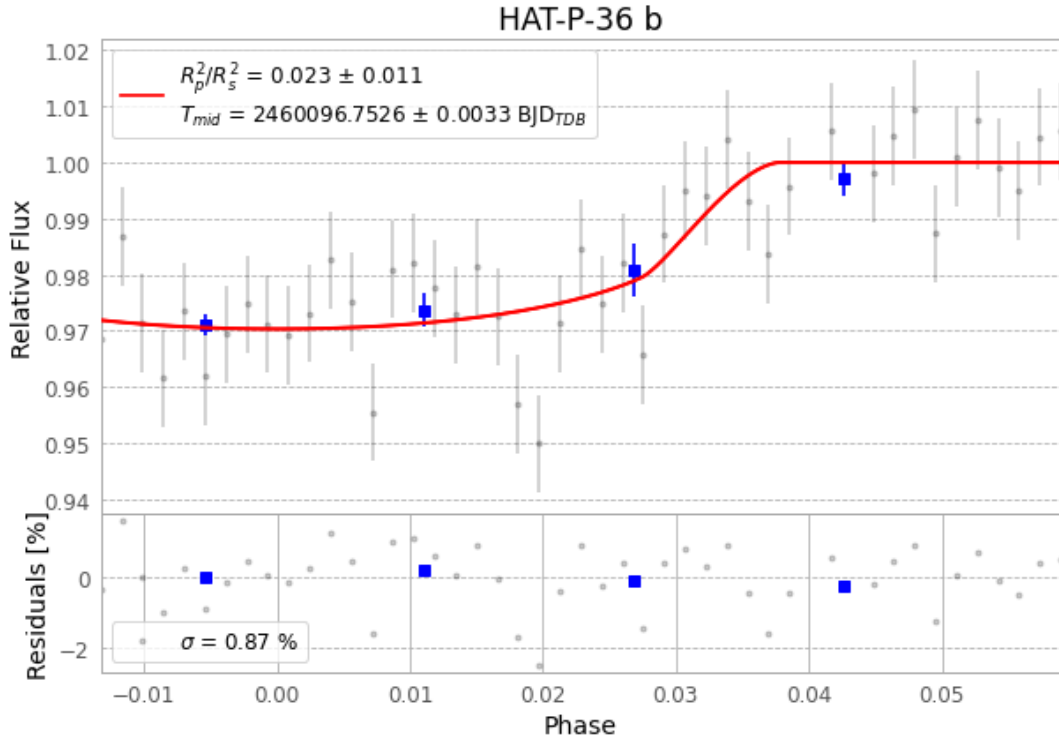


Figure 1 — FinalLightCurve_HAT-P-36 b_2023-05-31.png (SHA-256 7b8be40fd0b85192...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [258, 412], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b6e4f76f833aa8f1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

47 FITS frames (31 MB), HATP-36230601053615.FITS → HATP-36230601075415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-230601.log (05bf8feb501d...), FinalLightCurve_HAT-P-36 b_2023-05-31.png (7b8be40fd0b8...), FinalLightCurve_HAT-P-36 b_2023-05-31.csv (1c525e371f7b...)

Compute resources

Wall time 130.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 12:47Z.